

ORBITAL TERRAIN PERCEPTION & HAZARD-AWARE TRAVERSABILITY MAPPING FOR LUNAR ROVER AUTONOMY

A thesis submitted in partial fulfillment of the requirements for
the award of the degree of

MASTER OF TECHNOLOGY

in

DATA ANALYTICS

By

HARSH KUMAR SINGH

(205224003)



**DEPARTMENT OF COMPUTER APPLICATIONS
NATIONAL INSTITUTE OF TECHNOLOGY
TIRUCHIRAPPALLI – 620015**

JANUARY 2026

BONAFIDE CERTIFICATE

This is to certify that the project titled **ORBITAL TERRAIN PERCEPTION & HAZARD-AWARE TRAVERSABILITY MAPPING FOR LUNAR ROVER AUTONOMY** is a bonafide record of the work done by

HARSH KUMAR SINGH

(205224003)

in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Data Analytics** of the **NATIONAL INSTITUTE OF TECHNOLOGY, TIRUCHIRAPPALLI**, during the year 2025–26.

DR. ASHWINTH J.

Guide

DR. S. DOMNIC

Head of the Department

Project Viva-voce held on _____

Internal Examiner

External Examiner

ABSTRACT

Autonomous navigation of planetary rovers in the lunar South Polar region is a challenging task due to extreme terrain conditions such as steep slopes, craters, high surface roughness, and permanently shadowed regions (PSRs). These regions are of significant scientific interest due to the possible presence of water ice, but they also pose serious risks to rover mobility and safety. Traditional navigation approaches rely on onboard sensors, which have limited visibility and reduced performance in shadowed and complex terrains, making reliable navigation difficult. In this thesis, a framework for **orbital terrain perception and hazard-aware traversability mapping** is proposed using multi-source data from the Lunar Reconnaissance Orbiter (LRO). High-resolution imagery from the Lunar Reconnaissance Orbiter Camera (LROC) is used to extract visual features such as texture and shadow information, while elevation data from the Lunar Orbiter Laser Altimeter (LOLA) is utilized to derive topographic features including slope and surface roughness. The datasets are preprocessed, aligned, and combined using **feature-level fusion** to obtain a unified representation of the lunar terrain. A weighted scoring model is employed to estimate terrain risk by integrating slope, roughness, texture, and shadow features. Hazardous regions are identified using thresholding, and the computed risk is transformed into a **traversability score** ranging from unsafe to safe regions. The generated traversability map serves as a cost map for path planning using the **A* algorithm**, which determines optimal paths by minimizing both distance and terrain risk while avoiding hazardous areas. Experimental results demonstrate accurate alignment of LROC and LOLA datasets, effective feature extraction, and realistic path generation within acceptable computation time. The proposed approach improves navigation safety and efficiency by leveraging orbital data, reducing dependency on onboard sensors, and enabling robust pre-mission terrain assessment for future lunar exploration missions.

ACKNOWLEDGEMENT

I would like to extend my gratitude to **DR. G. AGHILA**, Director, National Institute of Technology, Tiruchirappalli for having provided all the facilities to carry out this project work.

I would like to thank **DR. S. DOMNIC**, Head of the Department, Department of Computer Applications, National Institute of Technology, Tiruchirappalli for allowing me to avail the facilities of the department during project work.

I express my sincere thanks and gratitude to my guide **DR. ASHWINTH J.**, Assistant Professor, Department of Computer Applications, National Institute of Technology, Tiruchirappalli who directed me from time to time throughout the project with valuable guidance, constant encouragement, and suggestions.

My hearty thanks to all the members of the Project Coordination Committee, Department of Computer Applications, National Institute of Technology, Tiruchirappalli, for their valuable suggestions during the project reviews.

I am also thankful to the faculty, staff members, and research scholars of the Department of Computer Applications, National Institute of Technology, Tiruchirappalli, for their support and suggestions throughout the project.

Harsh Kumar Singh

(205224003)

TABLE OF CONTENTS

Title	Page No.
ABSTRACT	i
ACKNOWLEDGEMENT	ii
TABLE OF CONTENTS	iii
LIST OF TABLES	vi
LIST OF FIGURES	vii
CHAPTER 1 INTRODUCTION	1
1.1 General Introduction	1
1.2 Lunar Terrain Analysis and Multimodal Data Fusion	1
1.3 Problem Statement	2
1.4 Objectives and Contributions	2
1.5 Organization of the thesis	3
CHAPTER 2 LITERATURE REVIEW	4
2.1 Introduction	4
2.2 Lunar Terrain Analysis and DEM-Based Methods	4
2.3 Image-Based Terrain Analysis	4
2.4 Multi-Source Data Fusion for Terrain Analysis	5
2.5 Traversability Mapping and Path Planning	5
2.6 Summary	5
CHAPTER 3 DATASET	7
3.1 Introduction	7
3.2 Dataset Overview	7
3.3 LROC Image Data	7

3.4	LOLA DEM Data	8
3.5	Data Alignment	9
3.6	Feature Extraction	10
3.7	Preprocessing Summary	10
3.8	Dataset Characteristics	11
3.9	Summary	11
CHAPTER 4 METHODOLOGY		12
4.1	Introduction	12
4.2	System Architecture	12
4.3	Architecture Overview	12
4.4	Data Preprocessing and Alignment	13
4.5	Feature Extraction	14
4.5.1	Slope Estimation	14
4.5.2	Surface Roughness	14
4.5.3	Texture Analysis	15
4.5.4	Shadow Detection	15
4.6	Feature Fusion	15
4.7	Traversability Map Generation	16
4.8	Path Planning Using A* Algorithm	16
4.9	Algorithm Steps	16
4.10	System Implementation	17
4.11	Summary	17
CHAPTER 5 EXPERIMENTS AND RESULTS		18
5.1	Introduction	18
5.2	Experimental Setup	18
5.3	Intermediate Results	18
5.3.1	LROC Mosaic Generation	18
5.3.2	LOLA Alignment	19
5.3.3	Feature Fusion	19

5.4	Traversability Map Analysis	19
5.5	Path Planning Results	20
5.5.1	Generated Navigation Paths	20
5.6	Performance Evaluation	24
5.6.1	Path Evaluation Metrics	24
5.6.2	Path Evaluation Results	26
5.7	Discussion	27
5.8	Summary	27
CHAPTER 6	CONCLUSION AND FUTURE WORK	28
6.1	Conclusion	28
6.2	Limitations	28
6.3	Future Work	29
6.4	Summary	29
REFERENCES		30

LIST OF TABLES

4.4.1 Alignment Verification Between LROC and LOLA Datasets	14
5.6.1 Evaluation Metrics for Generated Paths	26

LIST OF FIGURES

3.3.1 LROC Image of Lunar Surface Showing Cratered Terrain	8
3.4.1 LOLA Digital Elevation Model Representing Terrain Elevation	9
3.5.1 Aligned and Cropped LOLA DEM Corresponding to LROC Region .	10
4.6.1 Feature Fusion Result Combining LROC Imagery and LOLA Elevation Data	15
5.4.1 Generated Traversability Map of Lunar Terrain (Green: Safe Regions, Red: Hazardous Regions)	19
5.5.1 Path 1: Generated safe path avoiding hazardous regions	21
5.5.2 Path 2: Navigation path demonstrating obstacle avoidance	21
5.5.3 Path 3: Optimal path balancing distance and safety	22
5.5.4 Path 4: Path generated across complex terrain	22
5.5.5 Path 5: Safe path avoiding crater regions	23
5.5.6 Path 6: Path generated with long-distance traversal	23
5.5.7 Path 7: Short-distance navigation around obstacle	24

CHAPTER 1

INTRODUCTION

1.1 General Introduction

Lunar exploration has become a major focus in space research due to its scientific importance and potential for future human habitation. The lunar South Polar region is of particular interest because it contains permanently shadowed regions (PSRs), which are believed to harbor water ice deposits. These resources are critical for enabling long-term lunar missions and in-situ resource utilization.

However, the terrain in the lunar South Pole is extremely challenging. It consists of steep slopes, cratered surfaces, uneven topography, and deep shadowed regions that receive little to no sunlight. These conditions pose significant risks to rover mobility and navigation. Autonomous rovers operating in such environments must be capable of identifying safe paths while avoiding hazardous regions.

Traditionally, rover navigation relies heavily on onboard sensors such as cameras and LiDAR. While these sensors provide local information, they suffer from limited look-ahead capability and reduced performance in shadowed regions. This creates a need for robust pre-mission terrain analysis using orbital data to assist in safe and efficient navigation.

1.2 Lunar Terrain Analysis and Multimodal Data Fusion

Conventional terrain analysis approaches often rely on a single source of data, such as elevation models or imagery. While Digital Elevation Models (DEMs) provide valuable information about slope and roughness, they lack surface texture and shadow details. On the other hand, optical imagery provides visual information but does not capture elevation variations accurately.

Recent advancements in planetary remote sensing have enabled the availability of multi-source datasets such as LROC imagery and LOLA DEM. These datasets provide complementary information about the lunar surface. By combining visual and topographic data, it is possible to achieve a more comprehensive understanding of terrain characteristics.

Feature fusion techniques allow the integration of multiple data sources into a unified representation. This improves the ability to detect hazardous regions such as steep slopes, rough terrain, and shadowed areas. Such multimodal approaches are essential

for developing reliable traversability maps for rover navigation.

1.3 Problem Statement

The core problem addressed in this thesis is:

Given orbital data of the lunar surface, including LROC imagery and LOLA elevation data, generate a hazard-aware traversability map and determine a safe and optimal path for rover navigation.

The key technical challenges include:

- Extracting meaningful terrain features such as slope, roughness, texture, and shadow regions,
- Effectively combining heterogeneous data sources into a unified representation,
- Accurately identifying hazardous regions in complex terrain,
- Generating optimal navigation paths that minimize both distance and terrain risk.

Existing approaches often rely on either elevation data or imagery alone and do not fully exploit the benefits of feature fusion. Additionally, traditional navigation methods may not adequately account for terrain risk during path planning.

1.4 Objectives and Contributions

The main objectives of this thesis are:

- To integrate LROC imagery and LOLA DEM data for comprehensive terrain analysis,
- To extract important terrain features such as slope, roughness, texture, and shadow,
- To develop a feature fusion approach for combining visual and topographic information,
- To generate a traversability map representing terrain safety,
- To implement A* algorithm for optimal and hazard-aware path planning.

The key contributions of this work are:

1. A **multi-source data fusion framework** that combines LROC imagery and LOLA DEM for improved terrain understanding.
2. A **hazard-aware traversability mapping approach** based on weighted feature integration of slope, roughness, texture, and shadow information.
3. An **efficient path planning strategy** using the A* algorithm that minimizes both distance and terrain risk.
4. A comprehensive experimental analysis demonstrating effective terrain mapping and realistic path generation in lunar environments.

1.5 Organization of the thesis

The remainder of this thesis is organized as follows:

- **Chapter 2** presents a literature review on lunar terrain analysis, path planning techniques, and data fusion methods.
- **Chapter 3** describes the datasets used, including LROC imagery and LOLA DEM, along with preprocessing and alignment techniques.
- **Chapter 4** explains the proposed methodology, including feature extraction, feature fusion, traversability map generation, and A* based path planning.
- **Chapter 5** presents the experimental results, analysis of traversability maps, and evaluation of path planning performance.
- **Chapter 6** summarizes the conclusions of this work and discusses potential future improvements.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Terrain analysis and autonomous navigation are critical components of planetary exploration missions. In the context of lunar exploration, especially in the South Polar region, navigation is highly challenging due to the presence of craters, steep slopes, rough terrain, and permanently shadowed regions (PSRs). Accurate terrain perception and hazard detection are essential for ensuring safe and efficient rover operations.

This chapter reviews relevant literature on lunar terrain analysis, digital elevation models, image-based terrain understanding, traversability mapping, and path planning techniques. The limitations of existing approaches motivate the need for integrating multi-source data and developing hazard-aware traversability mapping methods.

2.2 Lunar Terrain Analysis and DEM-Based Methods

Digital Elevation Models (DEMs), particularly those generated from the Lunar Orbiter Laser Altimeter (LOLA), have been widely used for analyzing lunar terrain. DEMs provide valuable information about elevation, slope, and surface roughness, which are essential for identifying hazardous regions.

Several studies have utilized slope and roughness thresholds to classify terrain as safe or unsafe for rover navigation. While DEM-based methods are effective in capturing topographic features, they lack visual information such as texture and shadow, which are crucial for understanding surface characteristics.

Moreover, DEM-based approaches often fail to capture small-scale surface variations and illumination conditions, especially in permanently shadowed regions, limiting their effectiveness in complex terrains.

2.3 Image-Based Terrain Analysis

High-resolution imagery from the Lunar Reconnaissance Orbiter Camera (LROC) has been extensively used for analyzing surface texture, albedo, and shadow regions. Image-based approaches are effective in identifying visual patterns such as rocks, craters, and shadowed areas.

Techniques such as texture analysis, edge detection, and intensity-based thresholding are commonly used to extract meaningful features from images. However, image-only approaches do not provide elevation information, making it difficult to estimate slope and terrain geometry accurately.

Additionally, variations in illumination conditions and shadows can affect the reliability of image-based methods, especially in polar regions where lighting is highly uneven.

2.4 Multi-Source Data Fusion for Terrain Analysis

To overcome the limitations of single-source approaches, recent research has focused on combining multiple data sources such as DEMs and optical imagery. Data fusion techniques enable the integration of complementary information, resulting in improved terrain understanding.

Feature-level fusion methods combine extracted features such as slope, roughness, texture, and shadow information into a unified representation. This allows for more accurate identification of hazardous regions compared to using individual datasets.

However, many existing fusion methods use simple combination strategies and do not fully exploit the relationships between different features. There is a need for more robust and adaptive fusion techniques that can effectively capture terrain characteristics.

2.5 Traversability Mapping and Path Planning

Traversability mapping is a widely used approach for evaluating terrain safety. It involves assigning a traversability score to each region based on terrain features such as slope, roughness, and obstacles. Regions with high risk are marked as unsafe, while smoother and flatter regions are considered safe.

Path planning algorithms such as A*, Dijkstra, and Rapidly-exploring Random Trees (RRT) are commonly used for navigation. Among these, the A* algorithm is widely preferred due to its efficiency and ability to find optimal paths.

In recent studies, traversability maps have been used as cost maps for path planning, enabling algorithms to avoid hazardous regions. However, many approaches do not adequately integrate multiple terrain features or consider the combined effect of visual and topographic information.

2.6 Summary

The reviewed literature highlights the following key observations:

- DEM-based methods effectively capture elevation and slope but lack visual information.
- Image-based approaches provide texture and shadow details but do not represent terrain geometry accurately.
- Multi-source data fusion improves terrain understanding but requires more robust integration techniques.
- Traversability mapping combined with path planning is effective, but existing methods often lack comprehensive feature integration.

These limitations emphasize the need for a unified framework that combines LROC imagery and LOLA DEM data, performs feature-level fusion, and generates hazard-aware traversability maps for safe and efficient lunar rover navigation.

CHAPTER 3

DATASET

3.1 Introduction

This chapter describes the datasets used for terrain analysis and traversability mapping in the proposed framework. The study utilizes multi-source orbital data obtained from the Lunar Reconnaissance Orbiter (LRO), which provides detailed information about the lunar surface.

The datasets include high-resolution imagery from the Lunar Reconnaissance Orbiter Camera (LROC) and elevation data from the Lunar Orbiter Laser Altimeter (LOLA). These datasets are complementary in nature, capturing both visual and topographic characteristics of the lunar terrain. The combination of these modalities enables comprehensive terrain perception and hazard detection.

3.2 Dataset Overview

The dataset used in this work consists of two primary sources:

- **LROC NAC Imagery:** Provides high-resolution grayscale images of the lunar surface.
- **LOLA Digital Elevation Model (DEM):** Provides elevation data for computing terrain features.

The study focuses on the **lunar South Polar region**, which is characterized by complex terrain, including craters, steep slopes, and permanently shadowed regions (PSRs).

3.3 LROC Image Data

LROC Narrow Angle Camera (NAC) images provide detailed visual information about the lunar surface. These images are used to extract features such as texture, albedo, and shadow regions.

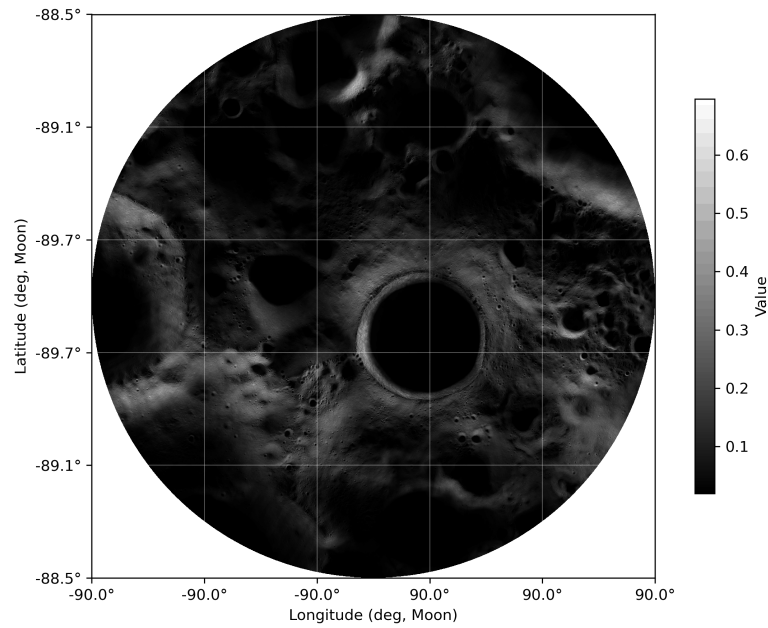


Figure 3.3.1: LROC Image of Lunar Surface Showing Cratered Terrain

The preprocessing steps for LROC images include:

- Image mosaicking to generate a continuous high-resolution map,
- Conversion to grayscale and normalization,
- Resizing and cropping to match the region of interest,
- Noise reduction and enhancement for better feature extraction.

These processed images help in identifying surface characteristics such as rocks, craters, and shadowed regions.

3.4 LOLA DEM Data

The LOLA Digital Elevation Model provides elevation measurements of the lunar surface. It is used to derive important terrain features such as slope, curvature, and surface roughness.

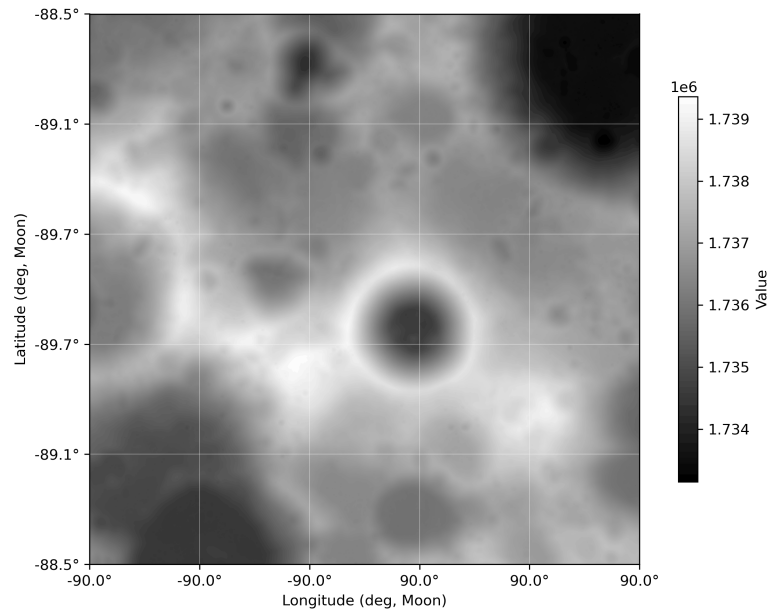


Figure 3.4.1: LOLA Digital Elevation Model Representing Terrain Elevation

The preprocessing steps for LOLA data include:

- Cropping the DEM to match the LROC image region,
- Resampling to ensure consistent resolution,
- Normalization of elevation values,
- Alignment with LROC imagery using coordinate reference systems.

The DEM plays a crucial role in identifying hazardous terrain based on elevation variations.

3.5 Data Alignment

Accurate alignment of LROC and LOLA datasets is essential for effective feature fusion. The datasets are aligned using spatial transformation techniques based on coordinate reference systems (CRS).

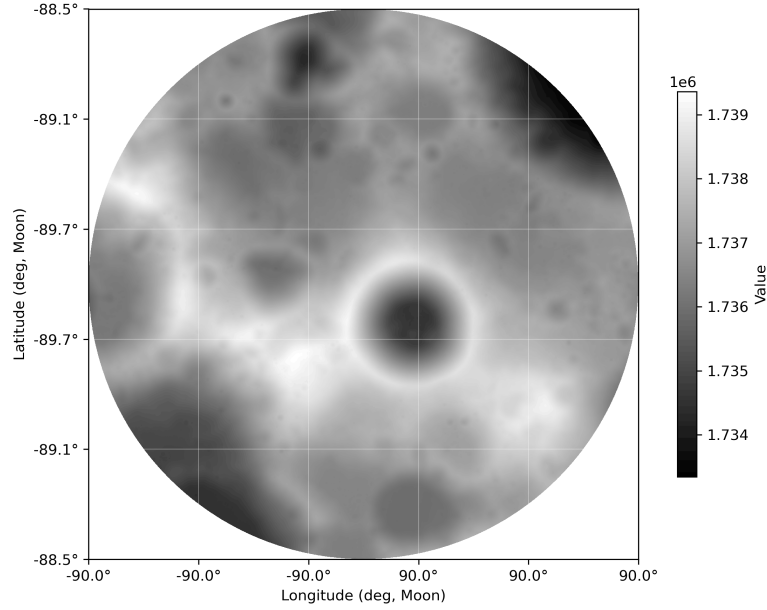


Figure 3.5.1: Aligned and Cropped LOLA DEM Corresponding to LROC Region

Normalized Cross-Correlation (NCC) is used to evaluate the alignment quality. A high NCC score indicates a strong correspondence between the datasets, ensuring that both visual and elevation features are correctly mapped.

3.6 Feature Extraction

The following features are extracted from the datasets:

- **Slope:** Derived from elevation gradients in the DEM.
- **Surface Roughness:** Calculated as the variation in elevation within a local neighborhood.
- **Texture:** Extracted from LROC images using local variance filters.
- **Shadow Regions:** Identified using intensity-based thresholding in images.

These features provide essential information for distinguishing between safe and hazardous terrain.

3.7 Preprocessing Summary

The final processed dataset consists of:

- **LROC Image Features** — representing texture and shadow information,
- **LOLA DEM Features** — representing slope and roughness,

- **Aligned Multi-Modal Data** — enabling feature-level fusion.

All features are normalized and prepared for integration in the traversability mapping stage.

3.8 Dataset Characteristics

The dataset exhibits several important characteristics:

- **High Terrain Variability:** Includes craters, slopes, and uneven surfaces.
- **Presence of Shadowed Regions:** Significant areas with low illumination.
- **Complex Topography:** Requires multi-source analysis for accurate understanding.

These characteristics highlight the need for combining visual and elevation data for effective terrain analysis.

3.9 Summary

This chapter presented an overview of the datasets used in this work, including LROC imagery and LOLA DEM. The preprocessing steps, alignment techniques, and feature extraction methods were discussed in detail. The next chapter describes the methodology used for feature fusion, traversability mapping, and path planning.

CHAPTER 4

METHODOLOGY

4.1 Introduction

The proposed framework focuses on orbital terrain perception and hazard-aware traversability mapping using multi-source lunar data. The methodology integrates high-resolution imagery from LROC and elevation data from LOLA to extract meaningful terrain features and generate safe navigation paths.

The approach consists of feature extraction, feature fusion, traversability map generation, and path planning using the A* algorithm. This chapter explains each component in detail.

4.2 System Architecture

The system architecture represents the complete workflow of the proposed framework, illustrating how multi-source data is processed to generate a safe navigation path. The architecture integrates both image-based and elevation-based inputs to perform terrain analysis.

The system follows a sequential pipeline where each stage processes the output of the previous stage. Initially, LROC imagery and LOLA DEM are preprocessed and aligned. These aligned datasets are then used for extracting terrain features such as slope, roughness, texture, and shadow information.

The extracted features are combined using a feature fusion module to generate a terrain risk map. This risk map is further converted into a traversability map, which serves as a cost map for path planning. Finally, the A* algorithm is applied to compute the optimal path that minimizes both distance and terrain risk.

4.3 Architecture Overview

The overall system processes two primary data sources—LROC imagery and LOLA DEM—to generate a traversability map and optimal navigation path. The major stages of the system are:

- Data preprocessing and alignment
- Feature extraction

- Feature fusion
- Traversability map generation
- Path planning using A*

4.4 Data Preprocessing and Alignment

The LROC and LOLA datasets are first preprocessed and aligned using coordinate reference systems (CRS). Accurate alignment ensures that each pixel in the image corresponds to the correct elevation value, which is essential for reliable feature extraction and fusion.

The preprocessing steps include cropping the datasets to a common region of interest, resampling to ensure consistent spatial resolution, and normalizing the data values. These steps ensure compatibility between the datasets before performing further analysis.

Normalized Cross-Correlation (NCC) is used to verify alignment quality. A high NCC score indicates strong correspondence between datasets, ensuring proper overlap between LROC imagery and LOLA elevation data.

To further validate the alignment between the LROC imagery and LOLA DEM, multiple spatial consistency checks were performed, including coordinate reference system (CRS) matching, shape consistency, resolution verification, affine transformation comparison, and bounding box alignment.

All alignment checks returned positive results, indicating perfect correspondence between the datasets. The CRS, spatial resolution, and affine transformation parameters were found to be identical, ensuring that both datasets are geometrically aligned without any distortion or mismatch.

Additionally, the origin shift in both X and Y directions was found to be zero, confirming that there is no spatial offset between the datasets. This perfect alignment ensures accurate pixel-to-pixel correspondence, which is essential for reliable feature fusion and traversability analysis.

Table 4.4.1: Alignment Verification Between LROC and LOLA Datasets

Parameter	Result
CRS Match	True
Shape Match	True
Resolution Match	True
Resolution Values	(5.0, 5.0)
Transform Match	True
Affine Transform	$\begin{bmatrix} 5.00 & 0.00 & -45490.00 \\ 0.00 & -5.00 & 45490.00 \\ 0.00 & 0.00 & 1.00 \end{bmatrix}$
Bounds Match	True
Bounding Box	(-45490, -45490, 45490, 45490)
Origin Shift (X, Y)	(0.0, 0.0)
NCC Score	0.7898
Alignment Status	Perfect Alignment

4.5 Feature Extraction

Terrain features are extracted from both datasets to characterize surface conditions.

4.5.1 Slope Estimation

Slope is calculated from the elevation data using gradient computation:

$$Slope = \tan^{-1} \left(\sqrt{\left(\frac{dz}{dx}\right)^2 + \left(\frac{dz}{dy}\right)^2} \right)$$

High slope values indicate steep regions that are unsafe for rover traversal.

4.5.2 Surface Roughness

Roughness is computed as the standard deviation of elevation values within a local window:

$$Roughness = \sigma(Z)$$

Regions with high roughness correspond to uneven terrain and obstacles.

4.5.3 Texture Analysis

Texture features are extracted from LROC images using local variance filters. These features help identify surface irregularities and patterns.

4.5.4 Shadow Detection

Shadow regions are detected using intensity-based thresholding. Low-intensity regions are classified as shadows, which may indicate hazardous or unknown terrain.

4.6 Feature Fusion

The extracted features are combined to generate a unified representation of terrain risk. A weighted scoring model is used:

$$Risk = w_1 \cdot Slope + w_2 \cdot Roughness + w_3 \cdot Texture + w_4 \cdot Shadow$$

where w_1, w_2, w_3, w_4 are weights assigned to each feature based on their importance. This fusion approach integrates both visual and topographic information for accurate hazard detection.



Figure 4.6.1: Feature Fusion Result Combining LROC Imagery and LOLA Elevation Data

The above figure illustrates the result of feature fusion, where both visual and

elevation information are combined to highlight terrain variations more effectively than individual datasets.

4.7 Traversability Map Generation

The risk values are converted into traversability scores:

$$Traversability = 1 - Risk$$

Values closer to 1 indicate safe regions, while values closer to 0 represent hazardous areas.

Thresholding is applied to classify regions:

- High slope and roughness $\rightarrow$ Unsafe
- Moderate values $\rightarrow$ Semi-safe
- Low values $\rightarrow$ Safe

The resulting traversability map acts as a cost map for path planning.

4.8 Path Planning Using A* Algorithm

The A* algorithm is used to compute the optimal path from start to goal positions. The cost function is defined as:

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$ is the accumulated cost from the start node,
- $h(n)$ is the heuristic (Euclidean distance to goal).

The algorithm selects paths that minimize both distance and terrain risk by incorporating the traversability map into the cost function.

4.9 Algorithm Steps

1. Initialize the open list with the start node
2. Select the node with the lowest cost
3. Expand neighboring nodes

4. Update cost based on traversability values
5. Repeat until the goal node is reached

This ensures efficient and safe navigation by avoiding hazardous regions.

4.10 System Implementation

The system is implemented using Python with libraries such as NumPy, OpenCV, and SciPy for image processing and numerical computations.

The processing pipeline includes:

- Image preprocessing and alignment
- Feature extraction
- Map generation
- Path planning visualization

The system is optimized to generate results within a short computation time.

4.11 Summary

This chapter presented the complete methodology of the proposed system, including data preprocessing, feature extraction, feature fusion, traversability mapping, and A* based path planning. The next chapter presents the experimental results and analysis.

CHAPTER 5

EXPERIMENTS AND RESULTS

5.1 Introduction

This chapter presents the experimental setup and results of the proposed framework for orbital terrain perception and hazard-aware traversability mapping. The evaluation focuses on the effectiveness of feature extraction, data alignment, traversability map generation, and path planning using the A* algorithm. The experiments are conducted using LROC imagery and LOLA DEM data from the lunar South Polar region. The results demonstrate the accuracy of terrain mapping and the efficiency of the path planning approach.

5.2 Experimental Setup

The system is implemented using Python and executed on a standard computing system. The following libraries are used:

- **NumPy** for numerical computations
- **OpenCV** for image processing
- **SciPy** for filtering and feature extraction
- **Matplotlib** for visualization

The experimental pipeline includes:

- Preprocessing and alignment of LROC and LOLA datasets
- Feature extraction (slope, roughness, texture, shadow)
- Feature fusion and traversability map generation
- Path planning using A* algorithm

5.3 Intermediate Results

5.3.1 LROC Mosaic Generation

A high-resolution mosaic of LROC images is generated to create a continuous representation of the lunar surface. This step ensures that the terrain is analyzed without discontinuities.

5.3.2 LOLA Alignment

The LOLA DEM is cropped and aligned with the LROC mosaic. The alignment accuracy is validated using Normalized Cross-Correlation (NCC), ensuring correct spatial correspondence.

5.3.3 Feature Fusion

The fused image combines visual and topographic information. This improves terrain understanding by integrating texture, elevation, and shadow features.

5.4 Traversability Map Analysis

The traversability map is generated using the weighted feature fusion approach.

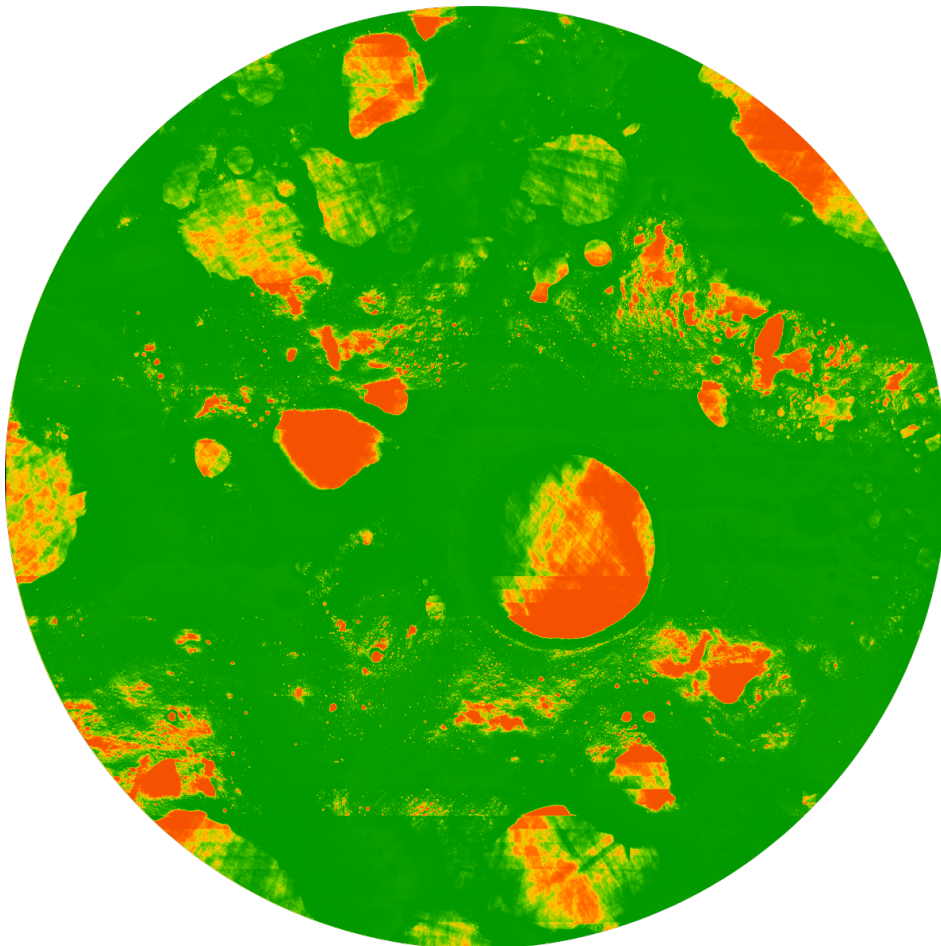


Figure 5.4.1: Generated Traversability Map of Lunar Terrain (Green: Safe Regions, Red: Hazardous Regions)

The above figure shows the generated traversability map, where green regions represent safe and highly traversable terrain, while red regions indicate hazardous areas such as steep slopes, rough surfaces, and crater boundaries. The map clearly

highlights safe navigation corridors and high-risk zones, which are crucial for effective path planning.

The map represents terrain safety as:

- **High traversability (safe regions)** — flat and smooth terrain
- **Low traversability (hazardous regions)** — steep slopes, rough areas, and shadowed regions

The map effectively highlights hazardous areas such as crater boundaries and steep slopes, enabling safer navigation.

5.5 Path Planning Results

The A* algorithm is applied to the traversability map to generate optimal paths. The algorithm successfully:

- Avoids high-risk regions such as craters and steep terrain
- Finds efficient paths between start and goal points
- Balances distance and terrain safety

The generated paths closely follow safe regions, demonstrating the effectiveness of the traversability-based cost function.

5.5.1 Generated Navigation Paths

To evaluate the robustness of the proposed path planning approach, multiple start and goal positions were selected across the traversability map. The A* algorithm was applied to generate optimal paths under different terrain conditions.

The color distribution observed in the path visualization represents different levels of terrain traversability. The **Yellow Regions** indicate highly traversable and safe areas corresponding to smooth and flat terrain where rover movement is efficient. The **Green Regions** represent moderately traversable areas where navigation is possible with minor caution. In contrast, the **Blue Regions** denote low traversability or hazardous terrain, typically associated with steep slopes, rough surfaces, or crater boundaries. It can be observed that the generated paths (shown in red) predominantly pass through yellow and green regions while avoiding blue regions, demonstrating the effectiveness of the hazard-aware path planning approach.

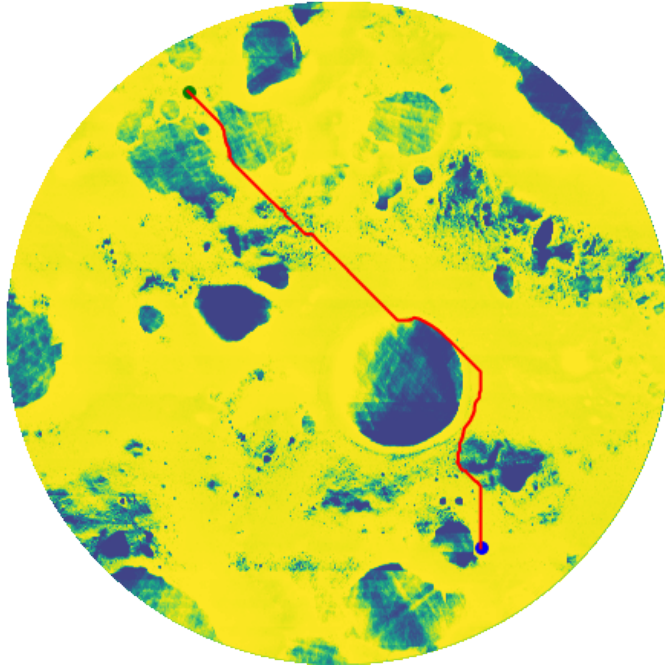


Figure 5.5.1: Path 1: Generated safe path avoiding hazardous regions

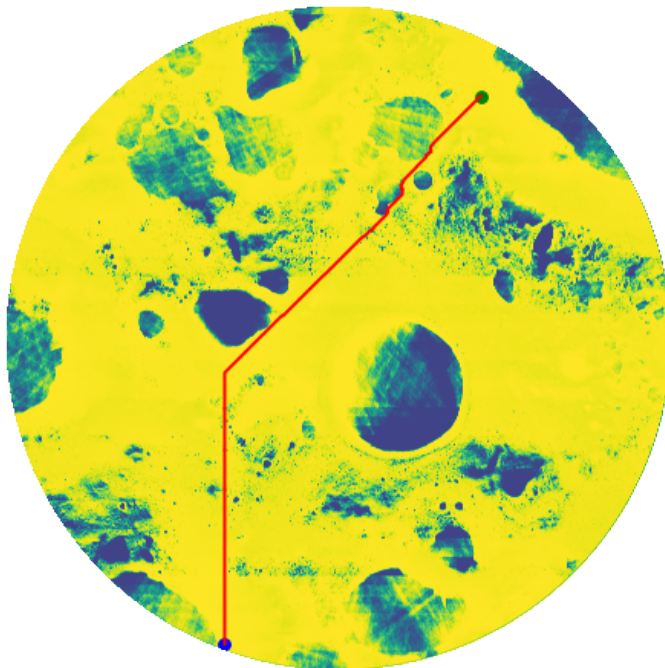


Figure 5.5.2: Path 2: Navigation path demonstrating obstacle avoidance

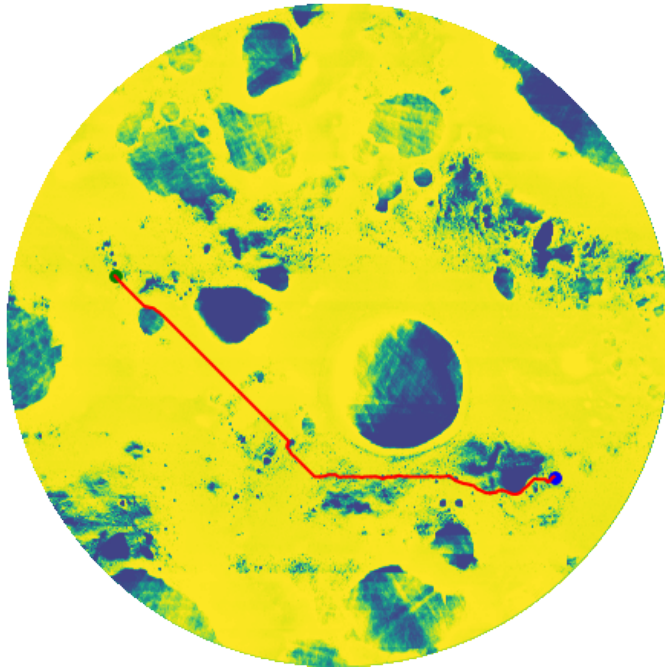


Figure 5.5.3: Path 3: Optimal path balancing distance and safety

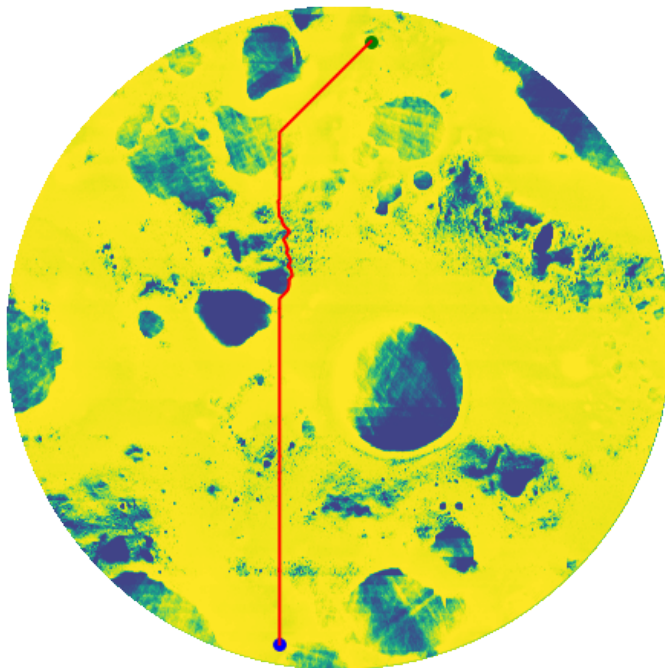


Figure 5.5.4: Path 4: Path generated across complex terrain

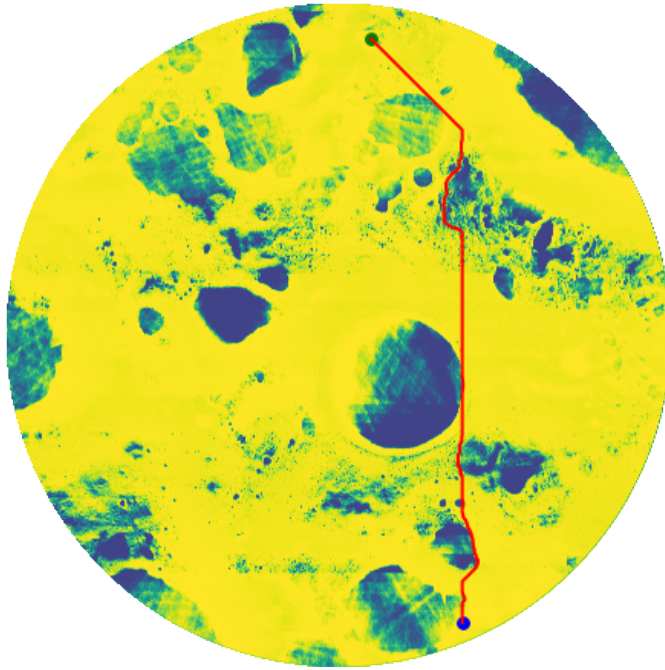


Figure 5.5.5: Path 5: Safe path avoiding crater regions

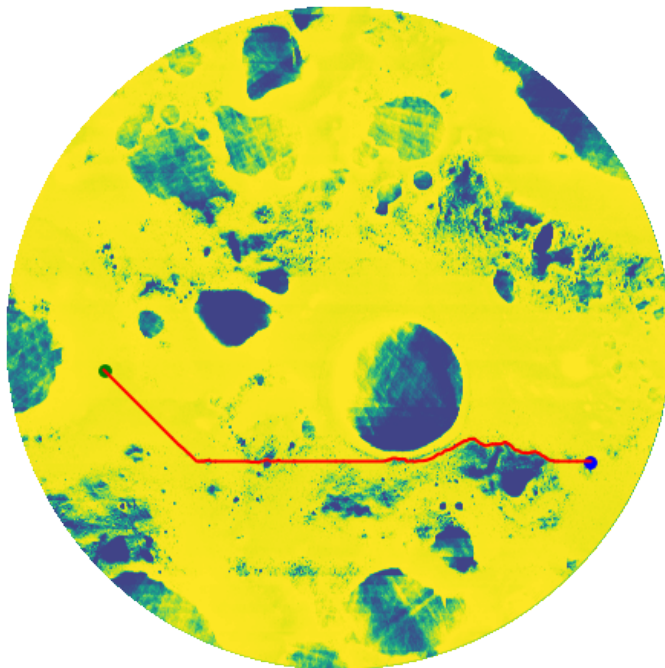


Figure 5.5.6: Path 6: Path generated with long-distance traversal

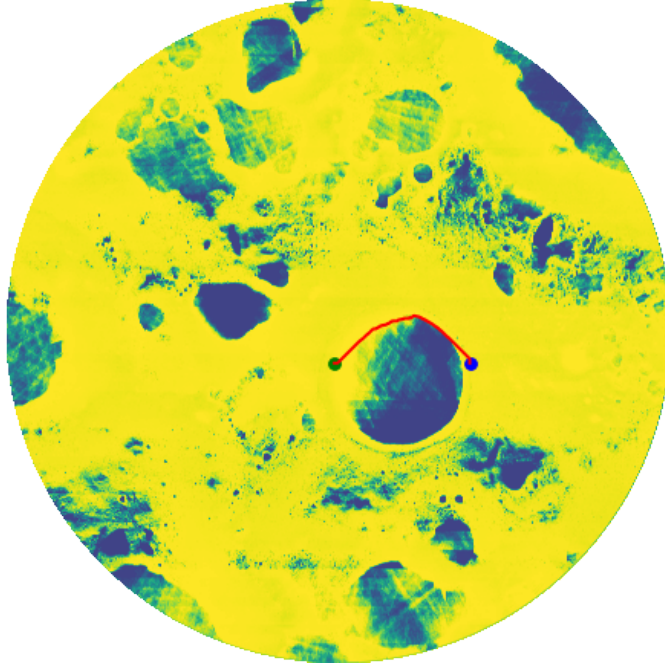


Figure 5.5.7: Path 7: Short-distance navigation around obstacle

The above results demonstrate that the proposed framework successfully generates safe and efficient paths under varying terrain conditions. The paths consistently avoid hazardous regions such as steep slopes and crater boundaries while maintaining reasonable path length.

5.6 Performance Evaluation

The performance of the system is evaluated based on:

- **Path Quality:** The generated paths avoid hazardous regions and follow safe terrain.
- **Computation Time:** The system generates results within 10–20 seconds.
- **Accuracy of Terrain Mapping:** The traversability map accurately reflects terrain conditions.

5.6.1 Path Evaluation Metrics

To quantitatively evaluate the quality of the generated paths, additional metrics are computed based on the traversability values along the planned path.

Let the path be represented as a sequence of points:

$$P = \{(r_1, c_1), (r_2, c_2), \dots, (r_n, c_n)\}$$

Cost Calculation

The cost at each point is defined as:

$$Cost_i = 1 - T(r_i, c_i)$$

where $T(r_i, c_i)$ represents the traversability value at that location.

Total Cost

$$Total\ Cost = \sum_{i=1}^n Cost_i$$

Average Cost

$$Average\ Cost = \frac{1}{n} \sum_{i=1}^n Cost_i$$

Average Traversability

$$Average\ Traversability = 1 - Average\ Cost$$

Interpretation of Metrics

The computed metrics are used to evaluate the quality of the generated path in terms of safety and efficiency.

- **Traversability:** A higher average traversability value indicates that the path passes through safer regions of the terrain. Ideally, the path should maximize traversability, meaning it avoids steep slopes, rough surfaces, and shadowed regions.
- **Cost:** Since cost is defined as the inverse of traversability, a lower cost corresponds to safer terrain. Therefore, the optimal path is expected to have minimum total and average cost. High cost values indicate that the path passes through hazardous regions.
- **Path Length:** The path length represents the number of steps required to reach the destination. While shorter paths are generally preferred, they may pass through unsafe regions. Hence, there is a trade-off between minimizing path length and maximizing safety.

In this work, the A* algorithm balances these factors by minimizing a combined cost function that accounts for both distance and terrain risk. As a result, the selected path may not always be the shortest, but it ensures safer navigation by avoiding hazardous regions.

An optimal path is therefore characterized by:

- High average traversability,
- Low total and average cost,
- Reasonable path length without compromising safety.

5.6.2 Path Evaluation Results

The performance of the generated paths is quantitatively evaluated using metrics such as path length, total cost, average cost, and average traversability. The results for different paths are summarized in Table 5.6.1.

Table 5.6.1: Evaluation Metrics for Generated Paths

Path	Path Length	Total Cost	Average Cost	Avg Traversability
Path 1	1155	25.9062	0.0224	0.9776
Path 2	1267	18.8397	0.0149	0.9851
Path 3	1033	28.5183	0.0276	0.9724
Path 4	1381	29.1399	0.0211	0.9789
Path 5	1364	31.6392	0.0232	0.9768
Path 6	1115	30.4719	0.0273	0.9727
Path 7	319	3.7685	0.0118	0.9882

From the results, it can be observed that all paths achieve high average traversability values (greater than 0.97), indicating that the generated paths predominantly pass through safe regions. Paths with lower total and average cost correspond to safer and more efficient routes.

Path 7 shows the highest traversability and lowest cost due to its shorter distance and minimal exposure to hazardous terrain. On the other hand, longer paths such as Path 5 and Path 6 have slightly higher costs due to increased traversal over varied terrain.

Overall, the results demonstrate that the proposed approach successfully balances path length and terrain safety, generating optimal and reliable navigation paths.

5.7 Discussion

The experimental results demonstrate that combining LROC imagery and LOLA DEM significantly improves terrain understanding. The feature fusion approach effectively captures both visual and topographic characteristics.

The A* algorithm, when combined with traversability mapping, produces safe and optimal navigation paths. The system overcomes limitations of traditional methods by integrating multiple features and reducing reliance on onboard sensors.

5.8 Summary

This chapter presented the experimental evaluation of the proposed framework, including intermediate results, traversability map analysis, and path planning outcomes. The results validate the effectiveness of the approach in generating safe and efficient navigation paths for lunar rovers.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This thesis presented a comprehensive framework for orbital terrain perception and hazard-aware traversability mapping for lunar rover autonomy. The proposed approach integrates multi-source data from the Lunar Reconnaissance Orbiter (LRO), combining high-resolution LROC imagery with LOLA Digital Elevation Models to achieve a detailed understanding of lunar surface conditions.

Unlike traditional methods that rely solely on onboard sensors or single-source data, the proposed framework leverages both visual and topographic information to extract critical terrain features such as slope, surface roughness, texture, and shadow regions. A feature-level fusion strategy was employed to integrate these heterogeneous features into a unified representation of terrain risk.

Based on the fused features, a traversability map was generated using a weighted scoring model that identifies hazardous and safe regions. This map was further utilized as a cost map for path planning using the A* algorithm. The resulting paths successfully avoided dangerous regions such as steep slopes and cratered areas, ensuring safe and efficient navigation.

Experimental results demonstrated accurate alignment of LROC and LOLA datasets, effective feature extraction, and realistic path generation within acceptable computation time. Overall, the findings confirm that combining orbital data with feature fusion and intelligent path planning significantly enhances rover navigation capabilities in complex lunar environments.

6.2 Limitations

Despite the effectiveness of the proposed framework, certain limitations exist:

- **Static feature weighting:** The weights used in feature fusion are predefined and may not adapt dynamically to different terrain conditions.
- **Limited real-time capability:** The system is designed for pre-mission planning and does not account for real-time environmental changes.
- **Dependence on data quality:** The accuracy of results depends on the quality and resolution of LROC and LOLA datasets.

These limitations highlight areas for potential improvement.

6.3 Future Work

Several directions can be explored to extend this work:

- **Machine learning integration:** Incorporating supervised or unsupervised learning methods to automatically learn terrain risk instead of using fixed weights.
- **Dynamic feature fusion:** Developing adaptive fusion techniques that adjust feature importance based on terrain characteristics.
- **Real-time navigation:** Integrating onboard sensor data with orbital maps for dynamic path planning.
- **Deep learning-based terrain classification:** Using convolutional neural networks for improved feature extraction and hazard detection.
- **3D terrain modelling:** Extending the approach to 3D representations for more accurate navigation.

These improvements can enhance the robustness and applicability of the system for future planetary exploration missions.

6.4 Summary

This thesis developed a robust and scalable approach for hazard-aware traversability mapping and path planning in lunar environments. By integrating multi-source orbital data and applying feature fusion techniques, the proposed method provides a reliable solution for safe rover navigation. The work contributes to the advancement of autonomous navigation systems and supports future lunar exploration and mission planning.

REFERENCES

- [1] Zhou et al., “A Safe and Efficient Global Path-Planning Method Considering Multiple Environmental Factors of the Moon,” *Remote Sensing*, 2025.
- [2] Leone et al., “A Candidate Location for the First Lunar South Pole Settlement,” *iScience*, 2023.
- [3] M. S. Robinson et al., “The Lunar Reconnaissance Orbiter Camera (LROC) Instrument Overview,” *Space Science Reviews*, 2010.
- [4] D. E. Smith et al., “The Lunar Orbiter Laser Altimeter Investigation on the Lunar Reconnaissance Orbiter Mission,” *Space Science Reviews*, 2010.
- [5] P. E. Hart, N. J. Nilsson, and B. Raphael, “A Formal Basis for the Heuristic Determination of Minimum Cost Paths,” *IEEE Transactions on Systems Science and Cybernetics*, 1968.
- [6] S. M. LaValle, “*Planning Algorithms*,” Cambridge University Press, 2006.
- [7] A. Angelova et al., “Learning Traversability of Terrain for Autonomous Ground Vehicles,” *ICRA*, 2007.
- [8] J. H. O’Neill and J. M. Thompson, “Surface Roughness Estimation for Planetary Rovers,” *Journal of Field Robotics*, 2014.
- [9] T. Blaschke, “Object Based Image Analysis for Remote Sensing,” *ISPRS Journal of Photogrammetry and Remote Sensing*, 2010.
- [10] H. Li, X. Shen, and J. Zhang, “Multi-Source Data Fusion in Remote Sensing: A Review,” *Information Fusion*, 2017.
- [11] J. P. Lewis, “Fast Normalized Cross-Correlation,” *Vision Interface*, 1995.
- [12] R. Volpe, “Rover Navigation Using Stereo Vision: Lessons from Mars,” *Journal of Field Robotics*, 1999.
- [13] A. Rankin et al., “Terrain-Based Navigation for Planetary Rovers,” *IEEE Aerospace Conference*, 2010.
- [14] J. Gallant and J. Wilson, “*Terrain Analysis: Principles and Applications*,” Wiley, 2000.